Preston **Rogers**

Autonomy / Robotics Engineer

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Education

Stanford University
Stanford, CA

M.S. in Electrical Engineering — *Sep 2019 - Jun 2021*

- Specialization: Control and Optimization
- Cumulative GPA: 4.1

University of California, Irvine
Irvine, CA

B.S. in Electrical Engineering — *Sep 2014 - Jun 2019*

- Specialization: Digital Signal Processing
- Cumulative GPA: 3.9 (Summa cum laude)

Relevant Academic Projects
Stanford, CA

Graduate School — *Sep 2019 - Jun 2021*

- Developed Newton-Raphson inverse kinematics solver for a 5-DOF EMG-controlled robot arm.
- Implemented collision-free path planning for a legged robot in MuJoCo using Sequential Convex Programming with linearized obstacle avoidance constraints.
- Trained a DNN to replicate an MPC trajectory planner for vehicle navigation.

Skills

Programming Languages

- C++, Python, C, MATLAB

Software & Tools

- ROS/ROS2, Docker, Git, Linux, MuJoCo, Gazebo, Altium Designer, PX4, GitLab CI/CD

Technical Domains

- Control Theory, Motion Planning, Autonomous Systems, Convex Optimization

Experience

NASA Jet Propulsion Laboratory
Pasadena, CA

Robotics Technologist — *Mar 2023 - Present*

- Developed a high-fidelity actuator model for the SRL arm for sample-tube motion validation.
- Lead controls software development for Superlimbs, supernumerary robotic limbs to assist astronauts on the moon.
- Leveraged CASAH software in design and development of multiple robotic platforms.
- Engineered and operated Extended Robust Aerial Autonomy platform, managing software infrastructure, telemetry GUI, and internal ROS1 nodes to ensure autonomous flight capabilities.
- Implemented tension-based PD velocity controller and level-winding system for TYMPO tethered drone spooling mechanism.
- Created software for operating flight EGSE for testing SRL flight actuators and delivered functional product to the SRL Lander Mechanical team and to engineers at Lockheed Martin space division.
- Performed longitudinal and lateral-directional stability analysis for DARPA EVADE VTOL aircraft in hover and fixed-wing trim configurations.

Dexterity AI
Redwood City, CA

Robotics Software Engineer — *Nov 2021 - Jan 2023*

- Developed convex shape estimation pipeline converting 3D RGBD point clouds to 2D package footprints for autonomous grasp planning.
- Implemented multi-strategy pick recovery procedures with gripper reorientation and translational offset fallbacks for failed grasps.
- Increased C++ operational space control test coverage by 20% using GoogleTest, validating motion primitives in simulation.

Glidewell Dental
Irvine, CA

Electrical Engineering Intern — *Jun 2019 - Sep 2019*

- Programmed an embedded microprocessor for use in an X-Ray Scanner.
- Constructed the electrical system incorporating a PLC for a Curing Tunnel.

Prof. Al Faruque Lab (AICPS)
Irvine, CA

Paid Undergraduate Researcher — *Feb 2018 - Mar 2019*

- Pioneered a C# GUI for real-time monitoring of sensor values.

Waterborne Skateboards
Irvine, CA

Co-founder and CAD Design Engineer — *Jan 2017 - Jan 2019*

- Refined a skateboard swivel truck using Solidworks.
- Forged a partnership with members of the UCI Wayfinder Incubator.

Publications

- E. Ballesteros, P. Rogers, J. Jenkins, K. Carpenter, and H. H. Asada, “Design of Supernumerary Robotic Limbs for the Augmentation of Astronauts Performing Partial-Gravity Extra-Vehicular Activities (EVAs),” *The International Journal of Robotics Research*, 2026.
- M. Dolci, J. Bowkett, P. Bailey, A. Boettcher, J. Kim, P. Rogers, et al., “Robotics Capabilities Development for Mars Sample Return Transfer Activities,” *2025 IEEE Aerospace Conference*, pp. 1–25, 2025.
- P. Rogers et al., “A High Fidelity Actuator Model for Robotic Autonomy in Space and Beyond,” *in preparation*.

Awards

- NASA JPL Team Certificate Award for outstanding engineering development and demonstration for the ERNEST active suspension vehicle.
- NASA JPL Team Certificate Award for the successful development and demonstration of end-to-end Sample Tube and Lid Transfer using Force Regulation on the SRL STS Bradbury Testbed.